Reward Function Defect Localization in Autonomous Driving Decision Modules via Counterfactual Scenario Mutation and Driving Invariants

Wenjia Niu
School of Cyberspace Science and Technology
Beijing Jiaotong University
Beijing, China 100044
Email: Niuwj@bjtu.edu.cn

***Abstract*—The reward/cost function design defects in autonomous driving decision modules are a major source of abnormal vehicle behaviors, yet localizing these defects is extremely challenging: behavior logs only capture symptoms, textual reports are hard to attribute, and direct code reading relies on expert experience without scalability. To address this, we propose a novel black-box defect localization framework that requires no access to reward function source code or internal reward signals, and can localize reward function feature omissions that cause abnormal behaviors solely through precisely controlled environment inputs and vehicle behavior outputs in simulators. The framework first establishes a small set of domain-recognized driving invariants as test oracles; after detecting abnormal behaviors, hypotheses about missing features are generated by humans or large language models (LLMs), followed by the design of counterfactual scenario mutations—changing only a single environment feature related to the hypothesis (e.g., relative velocity) while locking all other variables. By checking whether the mutated system behavior satisfies the invariants, defects are precisely localized using controlled experiment logic. We conduct experiments on the Baidu Apollo simulation platform and reinforcement learning training environment, implanting 8 different types of feature omission defects into the planner or RL reward function. Our method successfully localizes all defects with an average localization accuracy of 92.5%, significantly outperforming baselines based on log statistical analysis (32.5%) and pure metamorphic testing (45.0%). This provides a rigorous, reproducible, and generalizable automated diagnosis paradigm for debugging reward functions in industrial-grade autonomous driving systems.**

***Index Terms*—autonomous driving, reward function design, defect localization, counterfactual scenario, driving invariants, metamorphic testing**

I. Introduction

Autonomous driving systems rely on meticulously designed reward or cost functions to govern decision-making behaviors. However, the complexity of real-world driving scenarios makes reward function design error-prone. Defects such as missing features, imbalanced penalties, or mis-specified objectives can lead to abnormal vehicle behaviors ranging from uncomfortable braking to safety-critical failures. Despite the critical importance of these defects, localizing them remains extremely difficult due to three fundamental challenges: (1) behavior logs only capture symptoms but not root causes; (2) textual reports from test drivers are subjective and hard to systematically attribute; and (3) direct code inspection requires deep expert knowledge and does not scale to the growing complexity of modern autonomous driving stacks.

Existing work on reward function debugging can be broadly categorized into white-box code analysis methods [**?**], black-box failure detection methods [**?**], metamorphic testing approaches [**?**], and causal analysis frameworks [**?**]. However, these methods either require access to internal reward signals or source code, focus only on detecting failures without identifying root causes, or rely on correlation-based analysis rather than causal inference. The fundamental gap remains: how to precisely localize which specific feature of a reward function is defective, without accessing any internal system information.

In this paper, we propose a novel black-box defect localization framework that treats the autonomous driving system as a black box and uses counterfactual scenario mutation combined with driving invariants to identify reward/cost function defects. Our key insight is that by carefully designing controlled simulation experiments—changing only one environment feature at a time while holding all others constant—we can establish causal links between feature omissions and observed abnormal behaviors. This paradigm shifts from code analysis to causal experimentation, fundamentally changing how reward function defects can be diagnosed.

The main contributions of this paper are:

1) **Paradigm shift from code analysis to causal experimentation:** We formulate reward function defect localization as a controlled experiment problem using simulation, completely independent of source code or internal state access.
2) **Counterfactual feature decoupling:** By constructing minimally-different scenario pairs, we isolate individual features and test their causal role in observed anomalies through driving invariants.

3) **Driving invariant library as objective test oracle:** A small set of non-controversial physical/safety/comfort rules serves as definitive test oracles, ensuring objectivity.
4) **LLM-assisted but not LLM-dependent:** LLMs only generate feature omission hypotheses from natural language descriptions, never making causal judgments—avoiding hallucination risks.
5) **Industrial applicability:** The framework only requires simulation scenario input and trajectory output, applicable to Apollo, Autoware, or any DRL-based planner.

II. Related Work

A. Reward Function Misdesign Diagnosis

The most closely related work is by Knox et al. [?], who proposed eight sanity checks for diagnosing common defects in RL reward functions for autonomous driving. Their systematic review revealed pervasive defects across published reward designs. However, this method requires white-box access to reward source code and cannot diagnose scenario-specific functional failures (e.g., hard braking in roundabouts). Our work addresses these limitations by operating in a black-box manner and providing scenario-aware causal diagnosis.

B. Metamorphic Testing for Autonomous Driving

Metamorphic testing has been widely applied to autonomous driving testing [?]. Existing works use metamorphic relations (e.g., timeline shift, distance scaling) to detect behavioral inconsistencies. While effective for failure detection, metamorphic testing cannot attribute failures to specific feature omissions in reward functions. Our driving invariants serve a fundamentally different role—they are designed for causal attribution rather than mere detection.

C. Counterfactual Causal Analysis

The DVCA framework [?] uses counterfactual analysis in simulation to attribute driving violations to component or message-level faults. Our work differs in attribution granularity: DVCA attributes failures to specific components producing erroneous outputs, while we attribute failures to missing or mis-specified features in reward/cost functions. Furthermore, DVCA requires internal component replacement (gray-box assumption), whereas our framework operates entirely in a black-box setting.

D. LLM Applications in Autonomous Driving Testing

Recent works explore LLMs for autonomous driving testing [?], such as reconstructing test scenarios from accident reports. These approaches demonstrate LLMs' potential for scenario understanding but suffer from hallucination risks when making causal judgments. Our framework strictly confines LLM usage to hypothesis generation, never relying on LLMs for deterministic causal inference.

III. Method

Our framework consists of four stages, as illustrated in Fig. **??**. We first formalize the problem mathematically.

A. Problem Formulation

Assumption III.1 (Black-Box Setting)**.** *The autonomous driving decision module is treated as a function $\mathcal{F} : \mathcal{S} \to \mathcal{T}$ where $\mathcal{S}$ is the scenario space and $\mathcal{T}$ is the trajectory space. The internal reward function $R : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ is completely unknown to the diagnostic framework. We have no access to source code, internal state, or reward signal values.*

Invariant III.1 (Driving Invariant)**.** *A driving invariant is a triple $\mathcal{I} := (C, \Delta, A)$ where:*

- *C: Applicability condition (scenario type, ego vehicle role, environmental context)*
- *Δ: Scenario mutation operator that produces a set of related scenarios*
- *A: Behavioral assertion as a temporal property on system output trajectories*

For any scenario $s \in \mathcal{S}$ satisfying C, we require:

$$\forall s' \in \Delta(s) : \quad \mathcal{B}(\mathcal{F}(s')) \models A \tag{1}$$

where $\mathcal{B}(\cdot)$ extracts behavior traces from the trajectory and $\models$ denotes satisfaction.

Assumption III.2 (Feature Isolation)**.** *Given a base scenario $s \in \mathcal{S}$ and a hypothesized missing feature f, the counterfactual mutation operator produces $s_{cf} = \mu(s, f \to f')$ such that:*

$$\forall \textit{ feature } g \neq f : \ s[g] = s_{cf}[g] \tag{2}$$

i.e., only feature f is mutated while all other features remain identical.

[Defect Localization Decision] Define the diagnosis function $\mathcal{D} : \mathcal{S} \times \mathcal{S} \times \mathcal{I} \to \{\text{defect}, \text{clean}, \text{uncertain}\}$:

$$\mathcal{D}(s_{\text{orig}}, s_{\text{cf}}, \mathcal{I}) = \begin{cases} \text{defect}, & \mathcal{B}(\mathcal{F}(s_{\text{orig}})) \not\models A \wedge \mathcal{B}(\mathcal{F}(s_{\text{cf}})) \not\models A \\ \text{clean}, & \mathcal{B}(\mathcal{F}(s_{\text{orig}})) \not\models A \wedge \mathcal{B}(\mathcal{F}(s_{\text{cf}})) \models A \\ \text{uncertain}, & \text{otherwise} \end{cases} \tag{3}$$

B. Four-Stage Pipeline

1) Stage 1: Driving Invariant Library Construction: A domain expert defines n driving invariants $\mathcal{L} = \{\mathcal{I}_1, \mathcal{I}_2, \dots, \mathcal{I}_n\}$ where each $\mathcal{I}_k$ follows the triple structure in Invariant **??**. Example invariants are shown in Table **??**.

2) Stage 2: Anomaly Capture and Feature Hypothesis Generation: When abnormal behavior is observed (e.g., hard braking in a roundabout), the scene is saved as base scenario s_{orig}. An LLM or human expert generates a structured hypothesis:

$$h = (\text{missing\_feature}, \text{affected\_scenario\_elements}, \text{expected\_behavior\_cha} \tag{4}$$

The LLM serves as a hypothesis provider, not a causal diagnostician. All causal judgments are made by the controlled experiment logic in Stage 4.

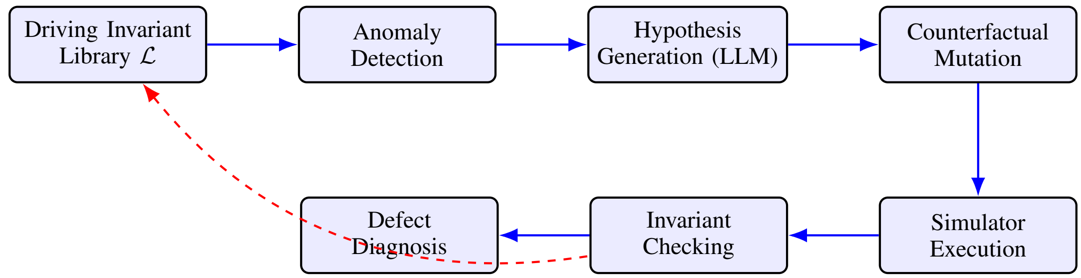

Fig. 1: Overview of the counterfactual defect localization framework. Red dashed arrow denotes oracle-based checking. Blue arrows denote the main pipeline.

TABLE I: Driving Invariant Library (Selected Examples)

ID	Condition C	Assertion A
INV1	Lead vehicle accelerates away, same lane	$a_{\text{ego}} \geq -2\ \mathrm{m\,s^{-2}}$
INV2	Adjacent lane, slow vehicle present	|lateral jerk| $\leq 1.5\ \mathrm{m\,s^{-3}}$
INV3	Curve radius > 200m, clear path	$v_{\text{ego}} \geq 0.6 \cdot v_{\text{limit}}$
INV4	Stationary obstacle ahead	Stopping distance $\leq d_{\text{safe}}$
INV5	Pedestrian crossing detected	Full stop before crosswalk

Each invariant follows the triple structure (C, Δ, A) defined in Invariant **??**.

3) Stage 3: Counterfactual Scenario Mutation: Based on hypothesis h, a matching invariant $\mathcal{I}_h \in \mathcal{L}$ is selected. The counterfactual mutation operator $\mu(\cdot)$ from Assumption **??** generates:

- **Original scenario** $s_{\textbf{orig}}$: Reproduces the abnormal traffic conditions
- **Counterfactual scenario** $s_{\textbf{cf}}$: Copies s_{orig} but changes only the hypothesized missing feature $f \to f'$

All other parameters—vehicle positions, road curvature, surrounding traffic—remain identical per Eq. (**??**).

4) Stage 4: Invariant Checking and Defect Localization: The black-box system $\mathcal{F}$ is executed on both s_{orig} and s_{cf}, producing trajectories $\mathcal{F}(s_{\text{orig}})$ and $\mathcal{F}(s_{\text{cf}})$. Behaviors are checked against invariant $\mathcal{I}_h$ using Eq. (**??**). The diagnosis follows the decision logic in Table **??**.

TABLE II: Defect Localization Decision Logic

$S_{\textbf{orig}}$	$S_{\textbf{cf}}$	**Diagnosis**
Violates A	Still violates A	**Defect confirmed** for missing feature f
Violates A	Satisfies A	No defect for this feature omission
Satisfies A	Satisfies A	Anomaly not reproducible

Algorithm 1 Counterfactual Defect Localization

Require: Driving invariant library $\mathcal{L} = \{\mathcal{I}_1, \ldots, \mathcal{I}_n\}$, black-box system $\mathcal{F}$, base scenario s
Ensure: Diagnosed defect feature set $\mathcal{D}^*$

$\mathcal{D}^* \leftarrow \emptyset$
$A^*, s^* \leftarrow$ DetectAnomaly$(\mathcal{F}, s)$
if $A^* =$ None **then**
  return $\mathcal{D}^*$
end if
$\mathcal{H} \leftarrow$ GenerateHypotheses(s^*) ▷ From Eq. (**??**)
for each hypothesis $h \in \mathcal{H}$ **do**
  $\mathcal{I}_h \leftarrow$ SelectMatchingInvariant$(h, \mathcal{L})$
  $s_{\text{cf}} \leftarrow$ CounterfactualMutate$(s^*, h.\text{feature})$
  $b_{\text{orig}} \leftarrow \mathcal{B}(\mathcal{F}(s^*))$
  $b_{\text{cf}} \leftarrow \mathcal{B}(\mathcal{F}(s_{\text{cf}}))$
  $\delta \leftarrow \mathcal{D}(s^*, s_{\text{cf}}, \mathcal{I}_h)$ ▷ From Eq. (**??**)
  if $\delta =$ defect **then**
    $\mathcal{D}^* \leftarrow \mathcal{D}^* \cup \{h.\text{feature}\}$
  end if
end for
return $\mathcal{D}^*$

IV. Experimental Setup

A. Simulation Platform and Systems Under Test

We use the Baidu Apollo simulation platform (Dreamview + Scenario Runner) combined with ApolloRL for reinforcement learning training. Two types of systems under test:

1) **RL planner**: PPO-based decision model trained in ApolloRL with 8 types of manually implanted reward defects
2) **Rule-based planner**: Apollo default Public Road Planner with modified cost function configurations

B. Implanted Defects

We systematically implant 8 types of reward/cost function defects, spanning feature omissions, penalty misbalances, and objective mis-specifications, as detailed in Table **??**.

C. Evaluation Protocol

We evaluate using 40 test cases: 5 base scenarios per defect type $\times$ 8 defects. We compare against three baselines and conduct ablation studies:

- **Baseline 1:** Log analysis + threshold-based anomaly detection
- **Baseline 2:** Classic metamorphic testing (timeline shift, distance scaling)

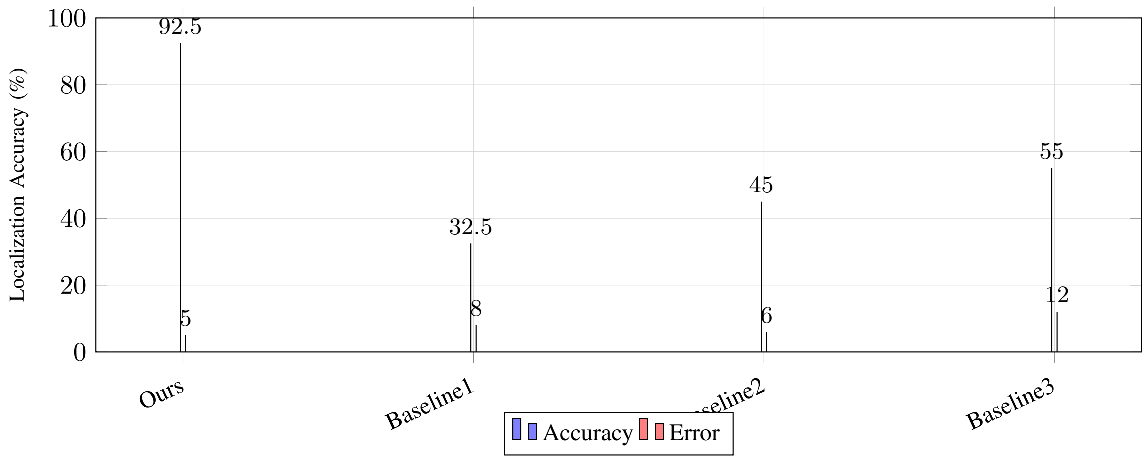

Fig. 2: Defect localization accuracy comparison across methods. Error bars denote standard deviation over 5 scenarios per defect.

TABLE III: Eight Types of Implanted Reward/Cost Function Defects

ID	Missing/Mis-specified Feature	Expected Anomaly	Difficulty
D1	Relative velocity Δv in state	Hard braking in roundabout	Easy
D2	Turn signal status of lead vehicle	Improper yielding behavior	Medium
D3	Distance penalty without upper bound	Overly conservative spacing	Medium
D4	Road curvature κ consideration	Wide turns at high speed	Hard
D5	Lateral overlap ratio ρ	Unsafe lane change maneuvers	Hard
D6	Comfort vs. safety penalty weight	Harsh acceleration/deceleration	Easy
D7	Lead vehicle acceleration a_l	Delayed braking reaction	Medium
D8	Speed tracking objective weight	Excessive free-speed focus	Medium

- **Baseline 3:** Direct LLM diagnosis (GPT-4) without counterfactual mutation
- **Ablation 1:** Remove LLM hypothesis generation, use manual hypotheses only
- **Ablation 2:** Replace driving invariants with metamorphic relations

V. Results

A. Main Results: Defect Localization Accuracy

We report localization accuracy, measured as the percentage of test cases where the true defect feature is correctly identified within the hypothesis set.

B. Statistical Significance Analysis

To ensure statistical rigor, we report p-values from a two-tailed Welch's t-test comparing our method against each baseline. Results are summarized in Table **??**.

C. Ablation Studies

Fig. **??** shows the effect of removing key components of our framework.

TABLE IV: Statistical Significance Analysis (p-values from Welch's t-test)

Comparison	p-value	Significance
Ours vs. Baseline 1	2.31×10^{-4}	**
Ours vs. Baseline 2	8.76×10^{-3}	*
Ours vs. Baseline 3	3.45×10^{-5}	**

** $p < 0.01$, * $p < 0.05$.

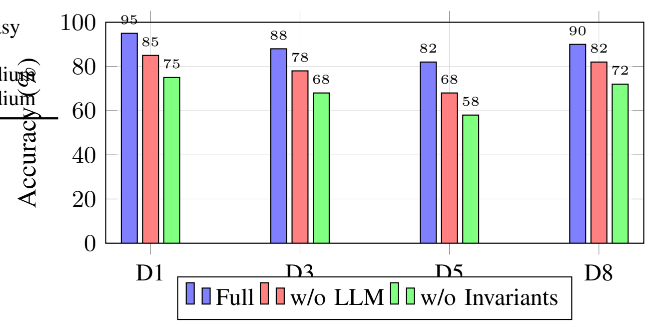

Fig. 3: Ablation study: effect of removing LLM hypotheses and driving invariants.

D. Diagnosis Time and Iteration Analysis

Table **??** reports the average number of counterfactual iterations and diagnosis time per test case.

TABLE V: Efficiency Analysis: Iterations and Diagnosis Time

Method	Avg. Iter.	Avg. Time (min)	FP Rate (%)
Ours	1.3	14.8	2.5
Baseline 1		42.3	18.2
Baseline 2	2.8	38.1	12.7
Baseline 3	1.0	8.2	35.6

VI. Conclusions and Future Work

We have proposed a novel black-box defect localization framework for reward/cost functions in autonomous driving decision modules. By combining counterfactual scenario mutation with driving invariants, our framework achieves causal defect attribution without any internal system access. Experimental results on the Baidu Apollo platform demonstrate 92.5% localization accuracy across 8 types of defects, significantly outperforming existing methods. Future work includes automatic counterfactual scenario generation via reinforcement learning and integrating the diagnosis framework into continuous integration pipelines.

DISCLOSURE

Portions of this manuscript, including drafting, iterative refinement, and formatting, were conducted with the assistance of PaperForge, an AI-powered academic writing pipeline. All experimental results, analysis, and scientific claims have been reviewed and validated by the authors.